



Ordinary Differential Equations

Complex numbers and Powers & Roots

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ENGR 213**

Lecture 8 (October 2, 2026)

Today's Lecture

- Complex numbers
- Powers and roots

Reference

Advanced Engineering Mathematics, D.G. Zill (7th edition), **Sections 17.1-17.2**

❑ Complex numbers

- A **complex number** is any number of the form $z = a + ib$, where a and b are real numbers and i is the imaginary unit.
- In $z = x + iy$, the real number x is called the real part of z ; the real number y is called the imaginary part of z .
- The real and imaginary parts of a complex number z are abbreviated **$Re(z)$** and **$Im(z)$** .
- A real constant multiple of the imaginary unit is called a **pure imaginary** number (e.g., $z = 6i$).
- Complex numbers $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ are equal, $z_1 = z_2$, if:
$$Re(z_1) = Re(z_2) \quad \text{and} \quad Im(z_1) = Im(z_2)$$
- A complex number $x + iy = 0$ if $x = 0$ and $y = 0$.

➤ Arithmetic operation on complex numbers

- For $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$:

- Addition: $z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$

- Subtraction: $z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2)$

- Multiplication: $z_1 \cdot z_2 = x_1x_2 - y_1y_2 + i(y_1x_2 + x_1y_2)$

- Division: $\frac{z_1}{z_2} = \frac{x_1x_2 + y_1y_2}{x_2^2 + y_2^2} + i\left(\frac{y_1x_2 - x_1y_2}{x_2^2 + y_2^2}\right)$

- The **modulus** or **absolute value** is given by $|z| = \sqrt{x^2 + y^2}$

- For $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ the following laws:
 - Commutative: $z_1 + z_2 = z_2 + z_1$ & $z_1 z_2 = z_2 z_1$
 - Associative: $z_1 + (z_2 + z_3) = (z_1 + z_2) + z_3$ & $z_1(z_2 z_3) = (z_1 z_2) z_3$
 - Distributive: $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$
- If z is a complex number, then the number obtained by changing the sign of its imaginary part is called the **complex conjugate** or, simply, the **conjugate** of z :

$$z = x + iy \longrightarrow \bar{z} = x - iy$$
 - If z is a real number, say $z = 7$, $\bar{z} = 7$.

- More laws for z and its conjugate $\bar{z}$:

$$\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$$

$$\overline{z_1 - z_2} = \bar{z}_1 - \bar{z}_2$$

$$\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$$

$$\overline{\begin{pmatrix} z_1 \\ z_2 \end{pmatrix}} = \begin{pmatrix} \bar{z}_1 \\ \bar{z}_2 \end{pmatrix}$$

- For $z = x + iy$:

$$z + \bar{z} = 2x$$

$$z\bar{z} = x^2 + y^2$$

$$\operatorname{Re}(z) = \frac{z + \bar{z}}{2}$$

$$\operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$$

$$z - \bar{z} = 2iy$$

EXAMPLE 1

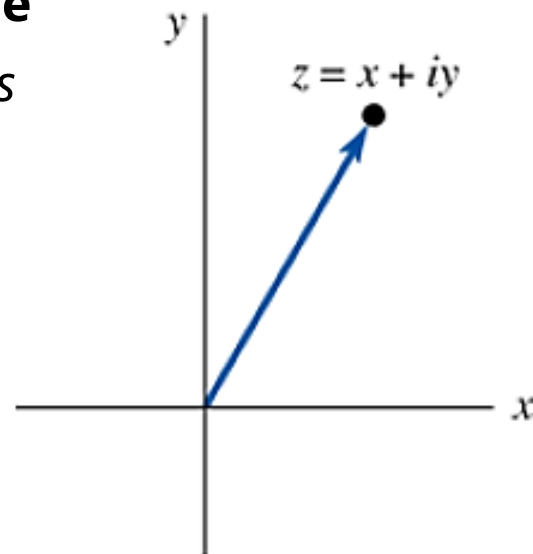
If $z_1 = 2 - 3i$ and $z_2 = 4 + 6i$, find z_1/z_2 and $1/z_1$.

* To divide z_1 by z_2 , we multiply both numerator and denominator of z_1/z_2 by the conjugate of z_2 .



➤ Geometric Interpretation

- A complex number $z = x + iy$ is uniquely determined by an ordered pair of real numbers (x, y) .
e.g., $z = 2 - 3i$ determines the ordered pair $(2, -3)$.
- So, complex number $z = x + iy$ is associated with a point (x, y) in a coordinate plane, and can also be viewed as a vector whose initial point is the origin and whose terminal point is (x, y) .
- This coordinate plane is called the **complex plane** or simply the **z-plane**: x-axis is called the *real axis* and y-axis is called the *imaginary axis*.
- The length of a vector z , or the distance from the origin to the point (x, y) , is $\sqrt{x^2 + y^2}$ or $|z|$.

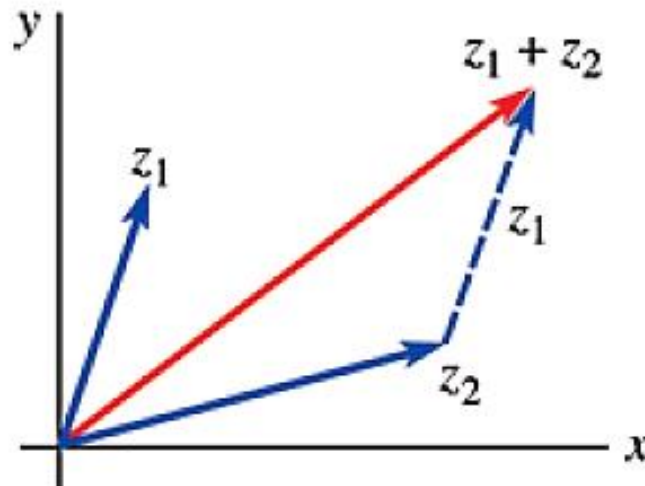


➤ Modulus of complex number and triangle inequality

- The sum of the vectors z_1 and z_2 is the vector $z_1 + z_2$.
- The length of the side of the triangle corresponding to the vector $z_1 + z_2$ cannot be longer than the sum of the remaining two sides:

$$|z_1 + z_2| \leq |z_2| + |z_1| \quad \text{triangle inequality}$$

$$|z_1 + z_2 + z_3 + \cdots + z_n| \leq |z_1| + |z_2| + |z_3| + \cdots + |z_n|$$



□ Powers and Roots

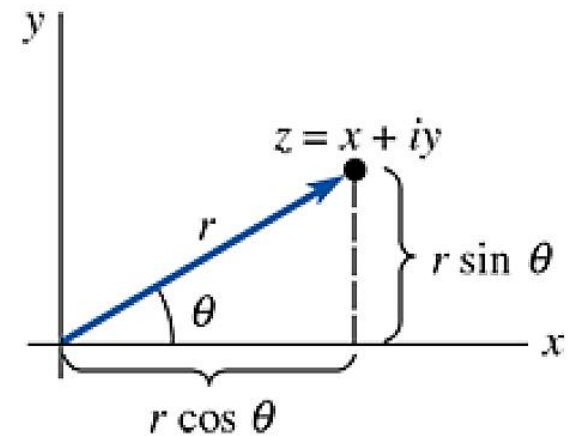
- **Polar form:** Rectangular coordinates (x, y) and polar coordinates (r, θ) are related by the equations $x = r \cos \theta$ and $y = r \sin \theta$, so:

$$z = x + iy \longrightarrow z = (r \cos \theta) + i(r \sin \theta)$$

polar form of z

$$r = |z|$$

$$\tan \theta = \frac{y}{x}$$



- The angle θ is called an argument of z , and is written $\theta = \arg z$.
 - * θ is measured in **radians** from the positive real axis, and is positive when measured counterclockwise and negative when measured clockwise.
 - * The argument of z in the interval $-\pi < \theta \leq \pi$ is called the principal argument.

EXAMPLE 2

Express $1 - \sqrt{3}i$ in polar form.



➤ Operation on complex numbers in polar form

- For $z_1 = (r_1 \cos \theta_1) + i(r_1 \sin \theta_1)$ and $z_2 = (r_2 \cos \theta_2) + i(r_2 \sin \theta_2)$:

- Multiplication: $z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$

- Division: $\frac{z_1}{z_2} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$

- $|z_1 z_2| = |z_1| |z_2|$ $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$

- $\arg(z_1 z_2) = \arg z_1 + \arg z_2$

- $\arg(z_1 / z_2) = \arg z_1 - \arg z_2$

➤ **Integer Powers of z :** $z^n = r^n(\cos n\theta + i \sin n\theta)$

➤ **De Moivre's Formula:**

When $z = (\cos \theta + i \sin \theta)$ $\longrightarrow (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$

➤ **Roots:** A number w is said to be an n th root of a nonzero complex number z if $w^n = z$.

If $z = r(\cos \theta + i \sin \theta)$ and $w = \rho(\cos \phi + i \sin \phi)$

$$w^n = z \longrightarrow \rho^n(\cos n\phi + i \sin n\phi) = r(\cos \theta + i \sin \theta)$$

$$w_k = r^{1/n} \left[\cos \left(\frac{\theta + 2k\pi}{n} \right) + i \sin \left(\frac{\theta + 2k\pi}{n} \right) \right]$$

$$k = 0, 1, 2, \dots, n-1$$

EXAMPLE 3

Find the fourth roots of $z = 1 + i$.